DS Lab 8: Data Visualization using Seaborn

Titanic Dataset

In [45]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style('whitegrid')
```

In [46]:
```
# Load Titanic dataset
dataset = sns.load_dataset('titanic')
dataset.head()
```

Out[46]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

In [47]:
```
dataset.info()
dataset.describe()
```

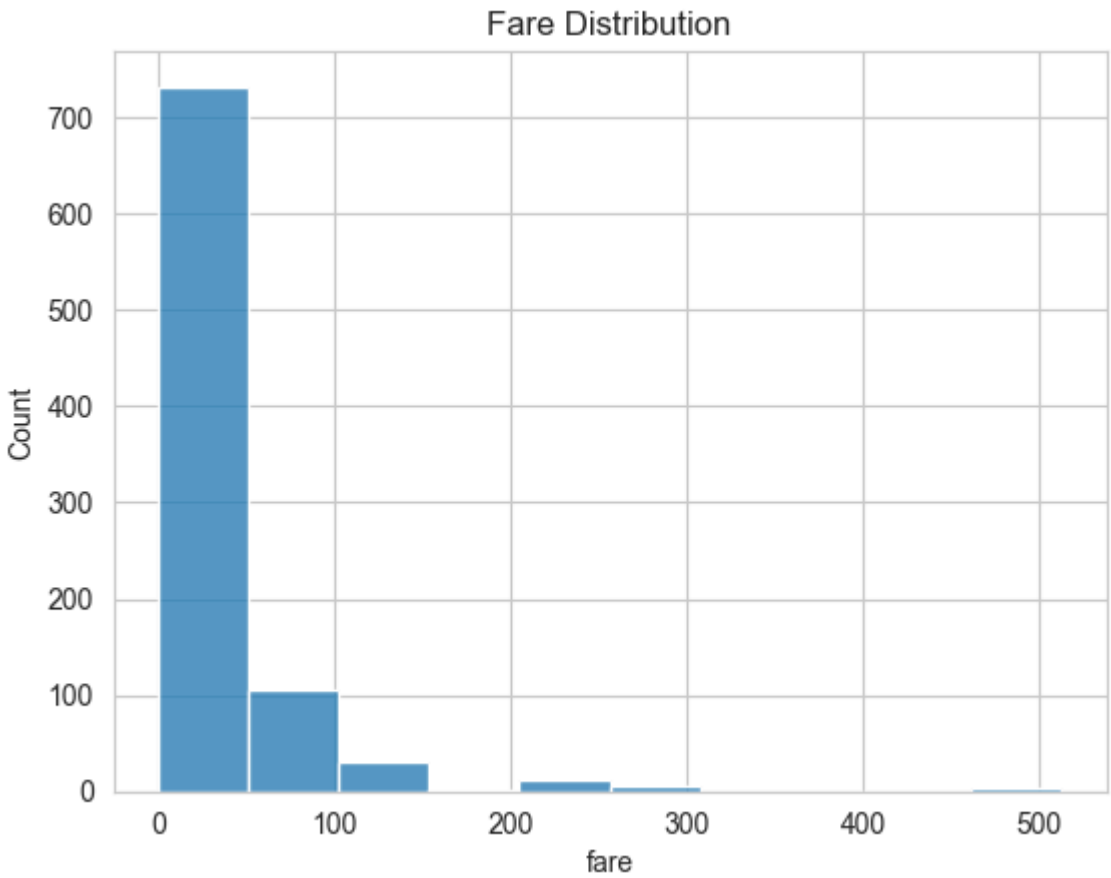
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

Out[47]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Histogram of Fare (Required)

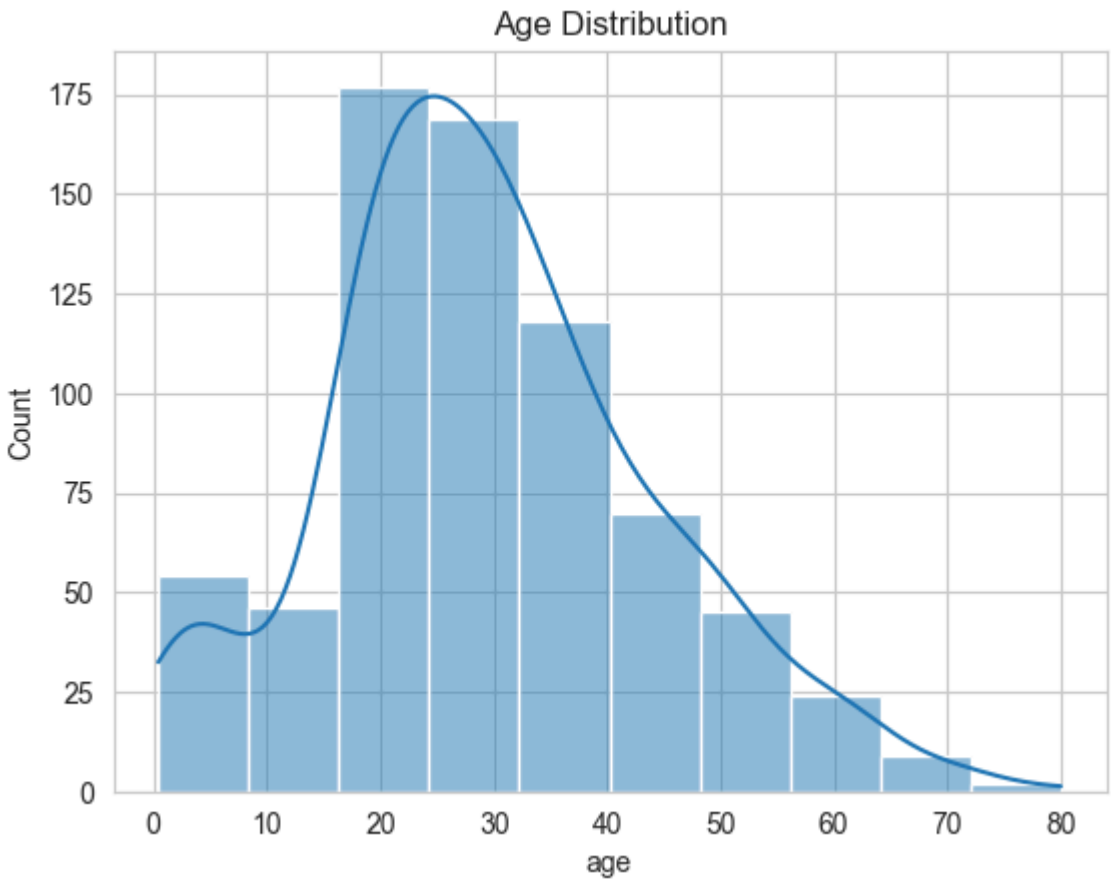
In [48]:
```
sns.histplot(dataset['fare'], bins=10, kde=False)
plt.title("Fare Distribution")
plt.show()
# Insight: Most passengers paid low fares, while only a few paid very high ticke
```



Distribution Plot

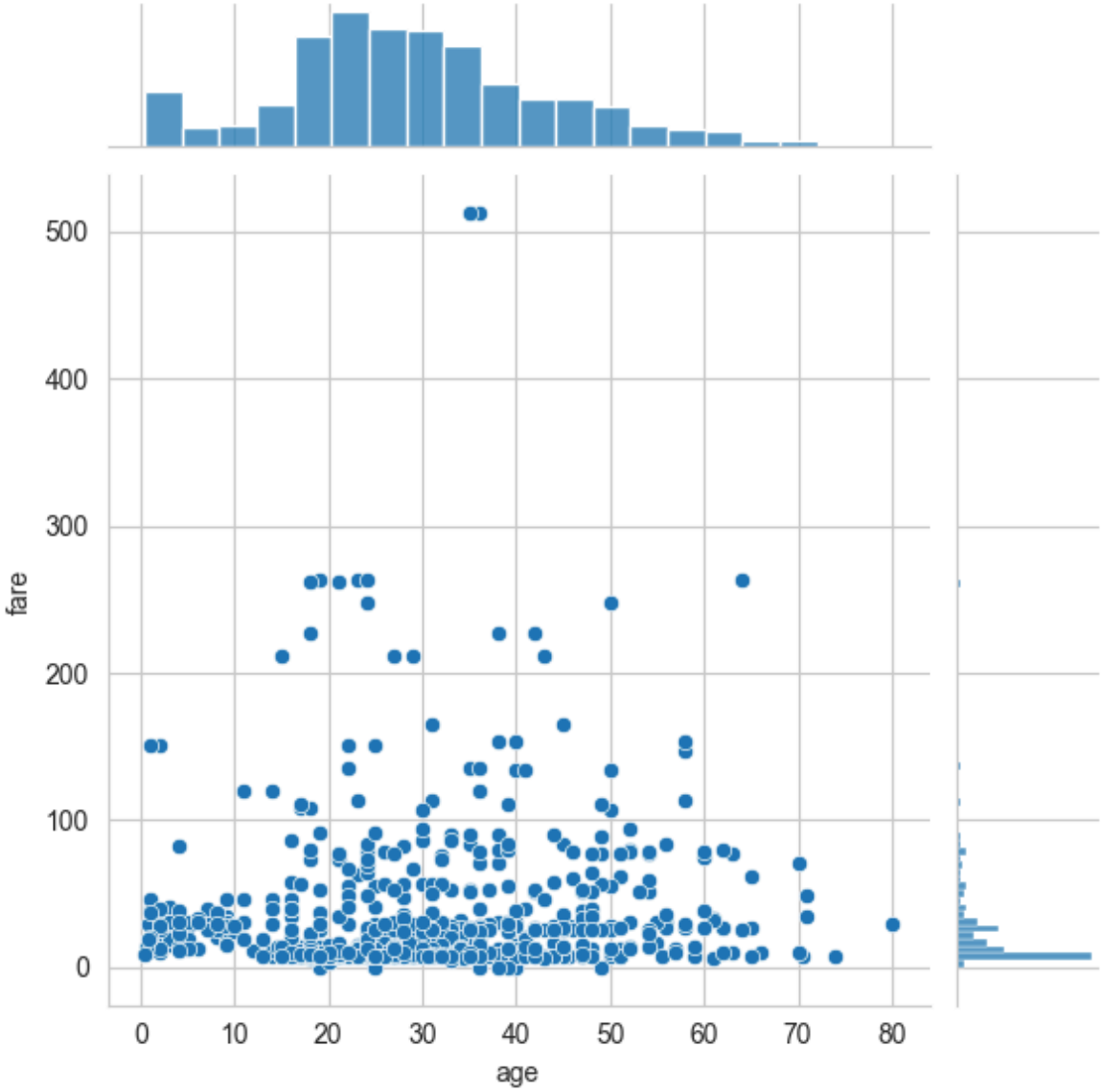
In [49]:
```
sns.histplot(dataset['age'], bins=10, kde=True)
plt.title("Age Distribution")
```

```
plt.show()
# Insight: Most passengers were young adults, with fewer very young children and
```



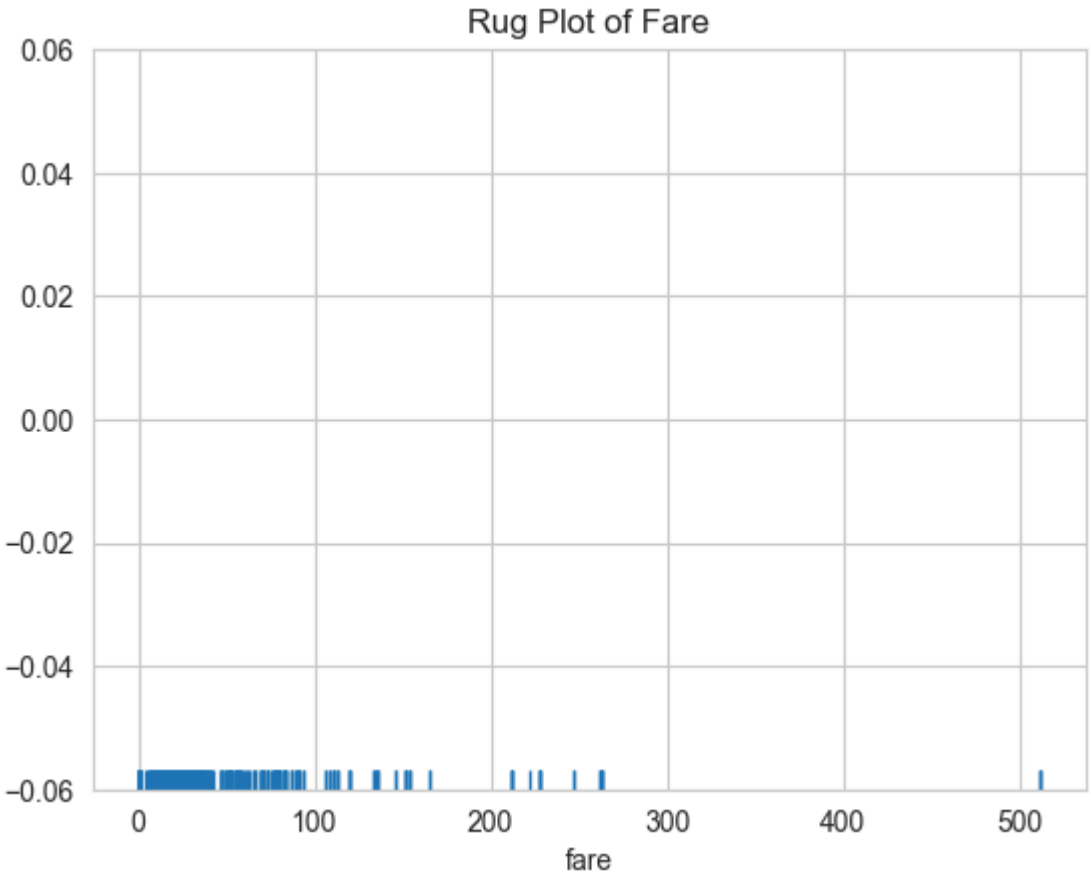
Joint Plot

In [50]:
```
sns.jointplot(x='age', y='fare', data=dataset, kind='scatter')
plt.show()
# Insight: Higher fares appear across adult ages, while many low fares are sprea
```



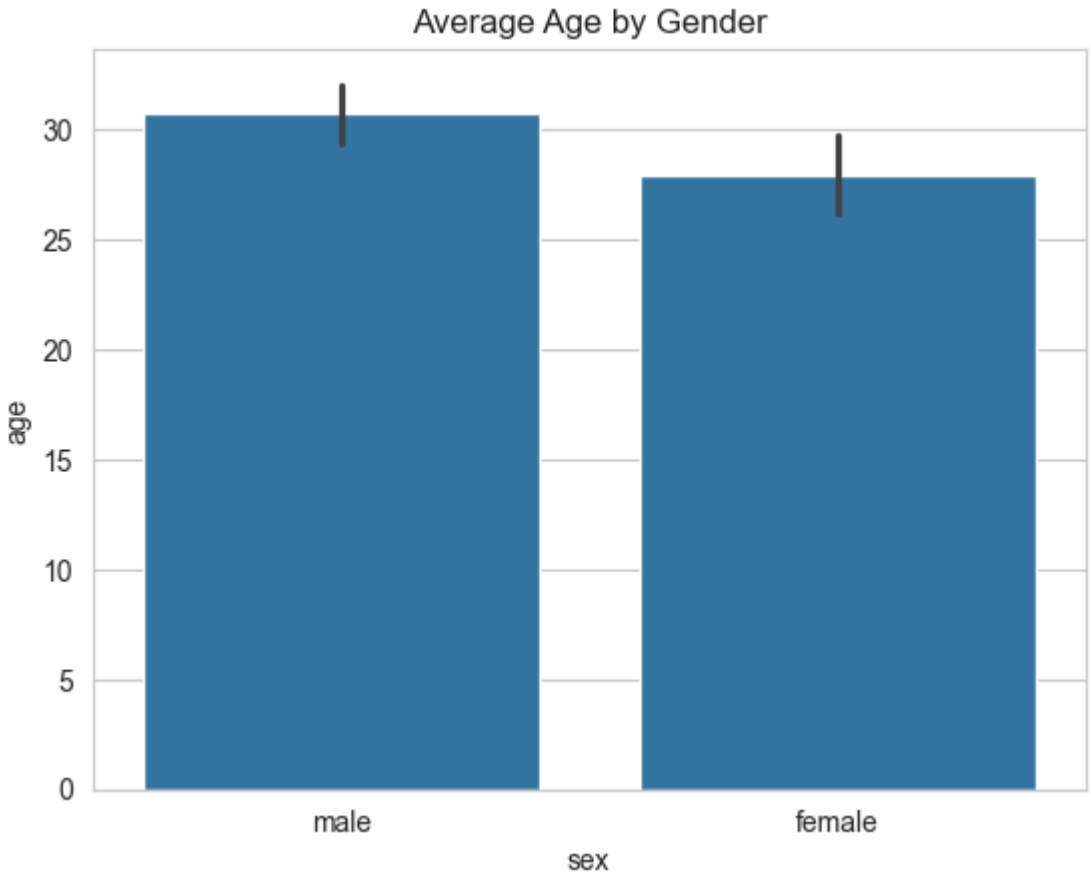
Rug Plot

In [51]:
```
sns.rugplot(dataset['fare'])
plt.title("Rug Plot of Fare")
plt.show()
# Insight: Fare values are densely packed at lower prices, showing few passenger
```



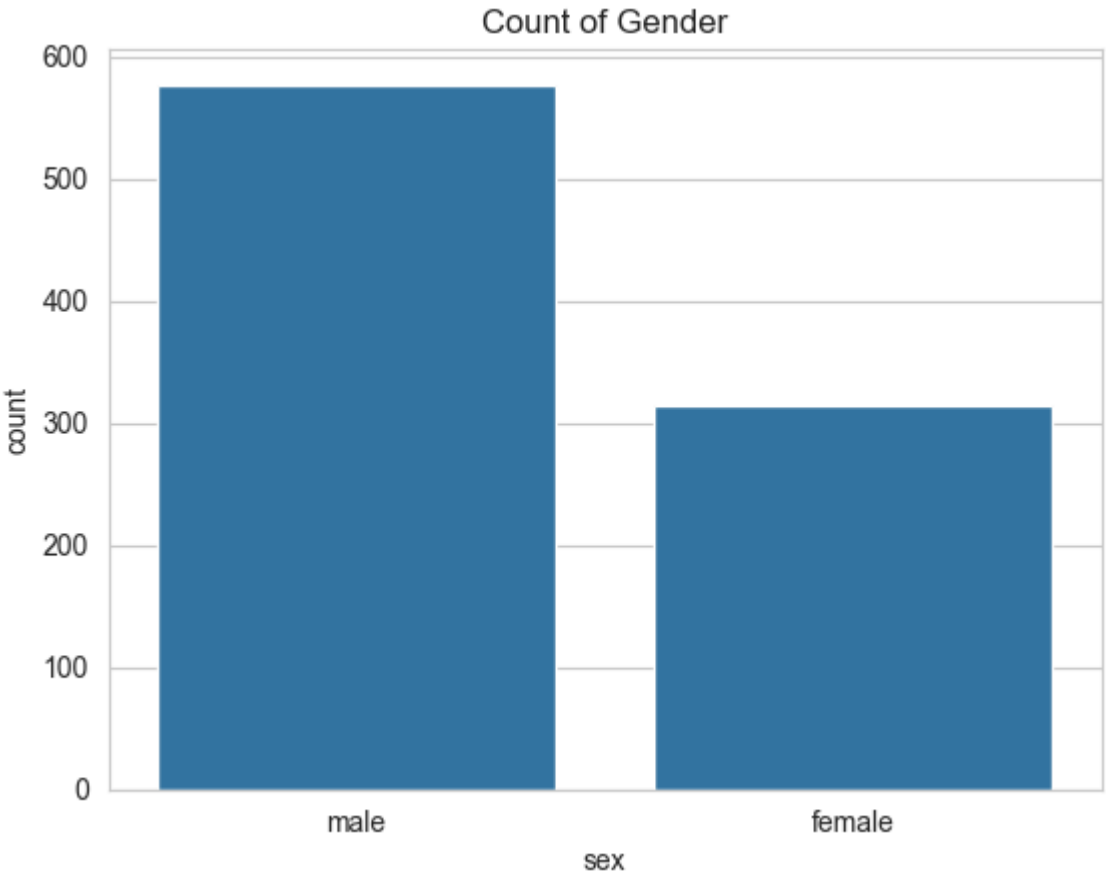
Bar Plot

In [52]:
```
sns.barplot(x='sex', y='age', data=dataset)
plt.title("Average Age by Gender")
plt.show()
# Insight: Male passengers have a slightly higher average age than female passen
```



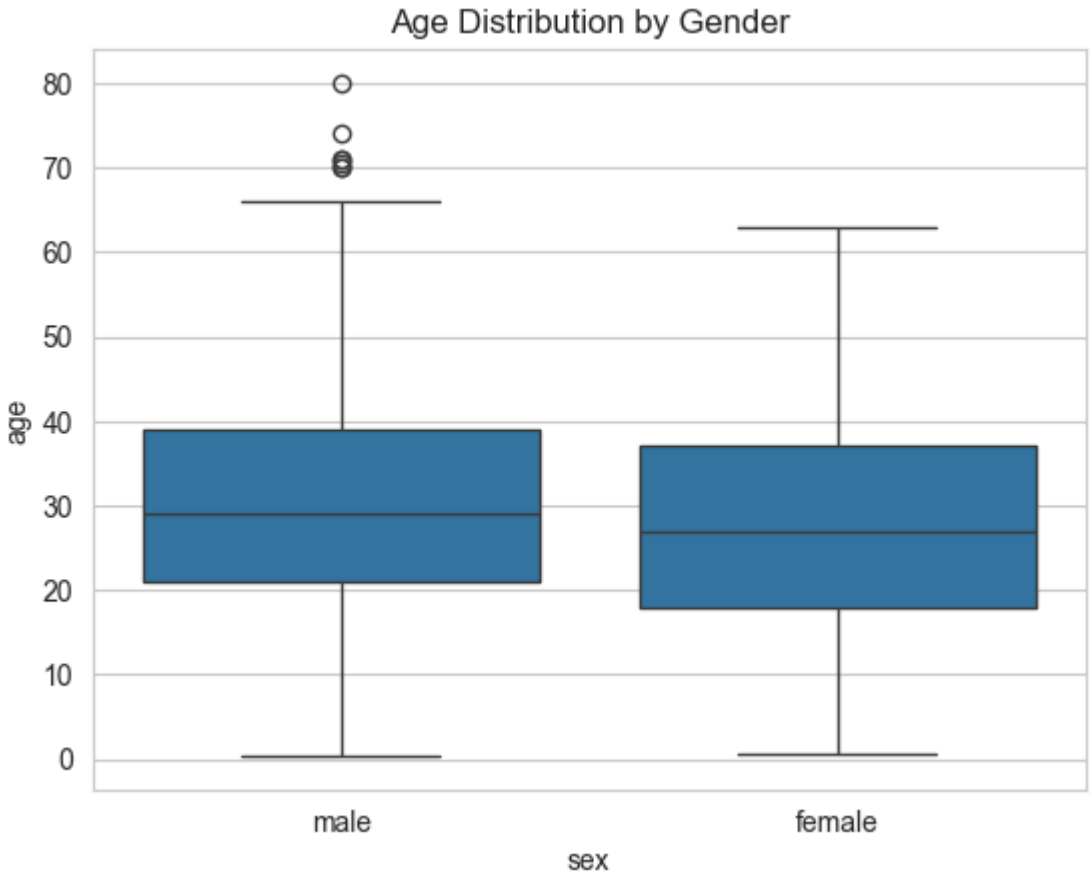
Count Plot

In [53]:
```
sns.countplot(x='sex', data=dataset)
plt.title("Count of Gender")
plt.show()
# Insight: Male passengers are more numerous than female passengers in this data
```



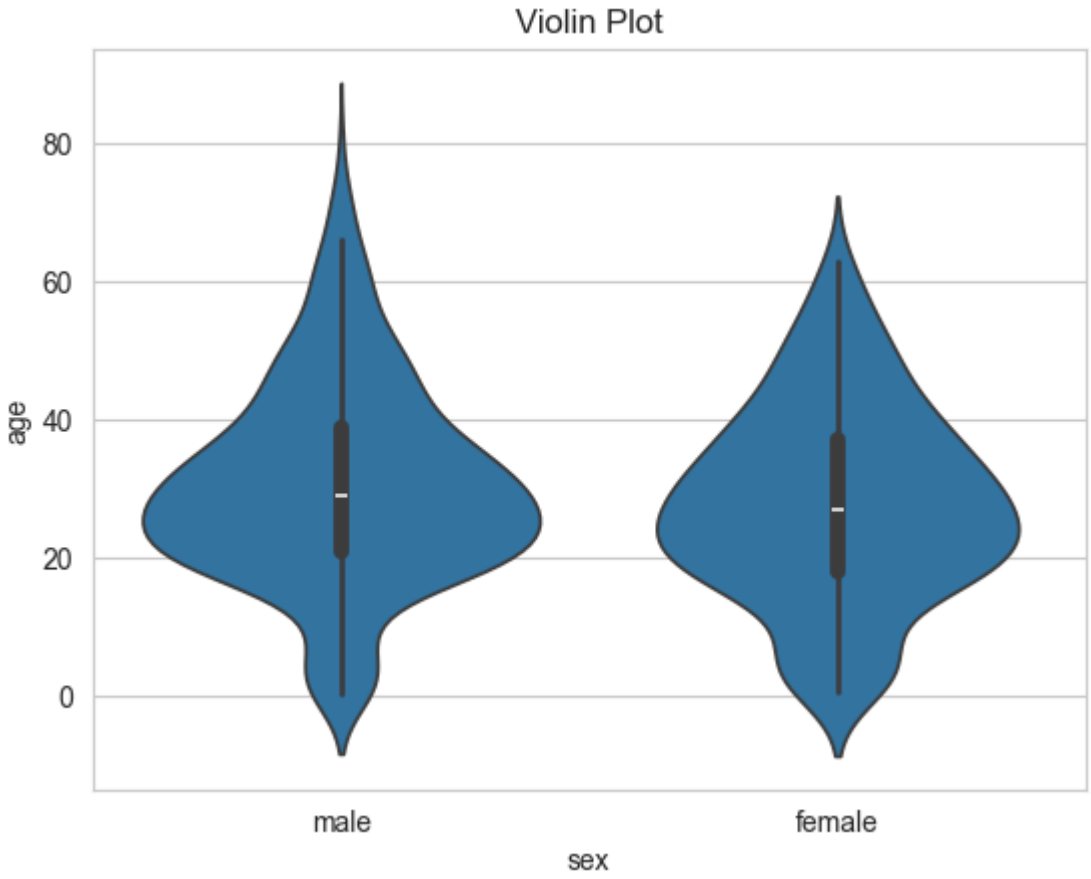
Box Plot

In [54]:
```
sns.boxplot(x='sex', y='age', data=dataset)
plt.title("Age Distribution by Gender")
plt.show()
# Insight: Male ages show a wider spread and more outliers than female ages.
```



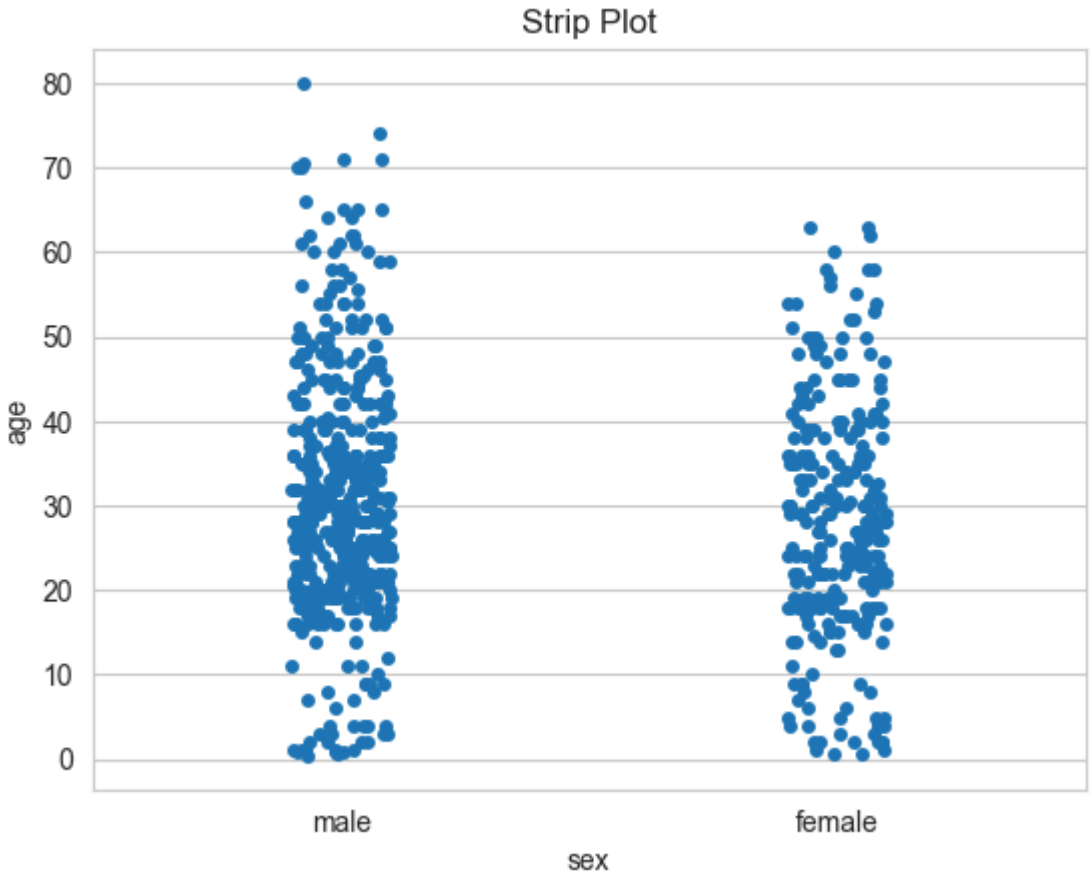
Violin Plot

In [55]:
```
sns.violinplot(x='sex', y='age', data=dataset)
plt.title("Violin Plot")
plt.show()
# Insight: Both genders center around young adult ages, with males showing broad
```



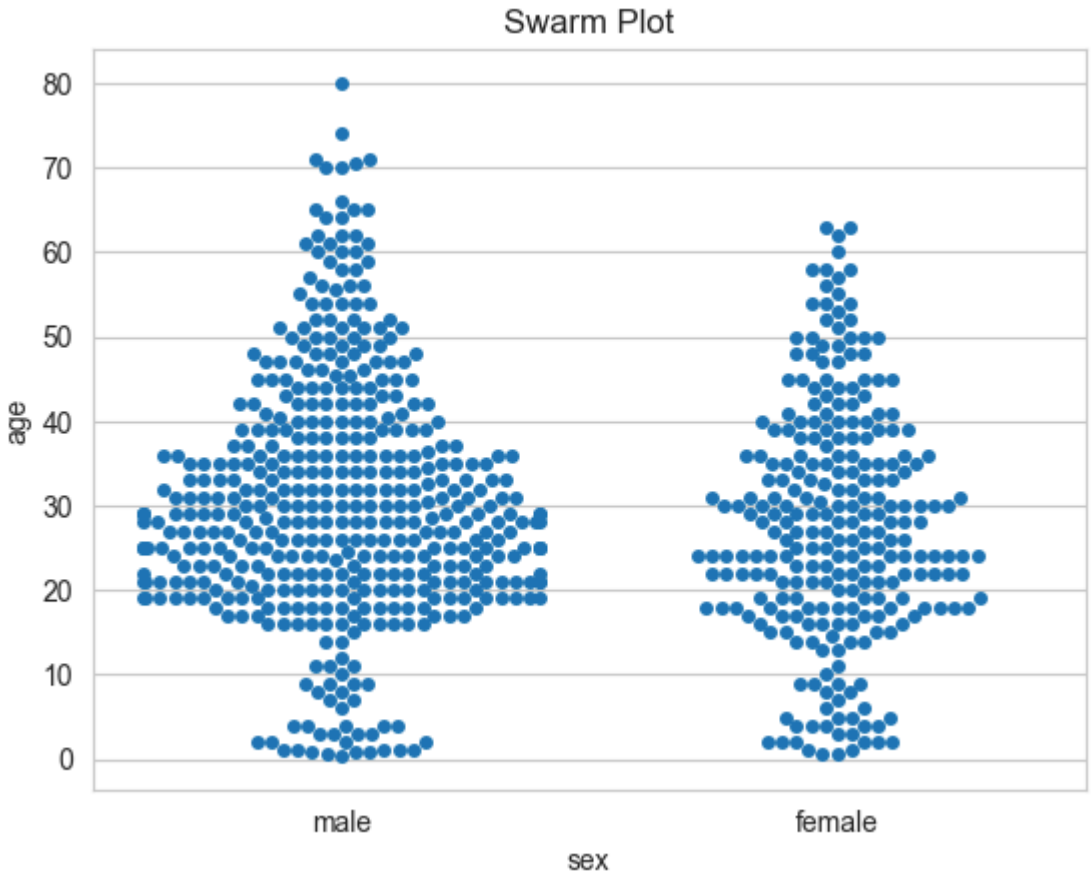
Strip Plot

In [56]:
```
sns.stripplot(x='sex', y='age', data=dataset, jitter=True)
plt.title("Strip Plot")
plt.show()
# Insight: Most points cluster below age 40, with overlap between male and femal
```



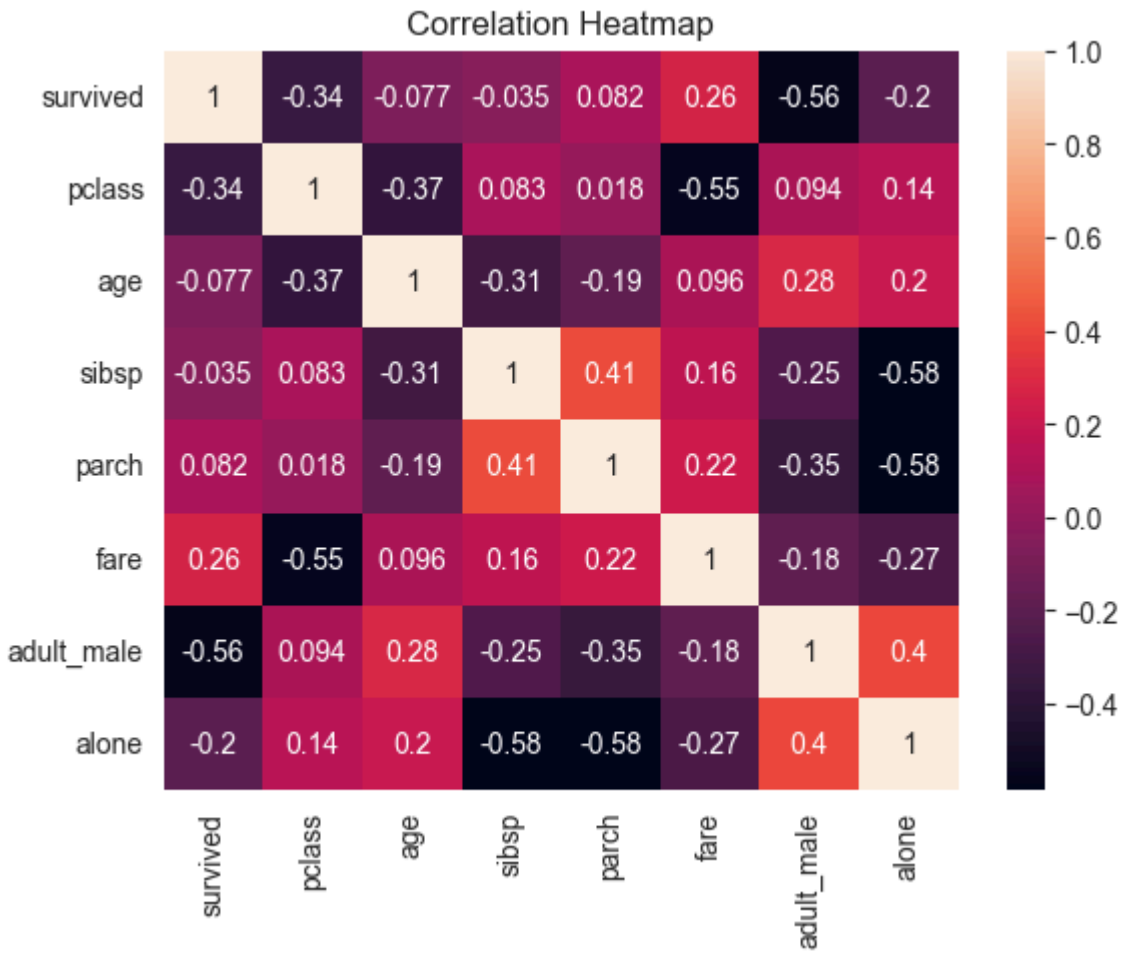
Swarm Plot

In [57]:
```
sns.swarmplot(x='sex', y='age', data=dataset)
plt.title("Swarm Plot")
plt.show()
# Insight: Age values overlap by gender, but males appear slightly more spread a
```



Heatmap

In [58]:
```
corr = dataset.corr(numeric_only=True)
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()
# Insight: Fare relates moderately with class-linked variables, while age shows
```


Correlation Heatmap
survived 1 -0.34 -0.077 -0.035 0.082 0.26 -0.56 -0.2
pclass -0.34 1 -0.37 0.083 0.018 -0.55 0.094 0.14
age -0.077 -0.37 1 -0.31 -0.19 0.096 0.28 0.2
sibsp -0.035 0.083 -0.31 1 0.41 0.16 -0.25 -0.58
parch 0.082 0.018 -0.19 0.41 1 0.22 -0.35 -0.58
fare 0.26 -0.55 0.096 0.16 0.22 1 -0.18 -0.27
adult_male -0.56 0.094 0.28 -0.25 -0.35 -0.18 1 0.4
alone -0.2 0.14 0.2 -0.58 -0.58 -0.27 0.4 1
survived pclass age sibsp parch fare adult_male alone
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4